

ECONOMIC RESEARCH PORTFOLIO

Sentiment, Shocks, and Speed of Adjustment

An ARDL–ECM Analysis of Market Sentiment, External Shocks, and Trading Activity on Indonesia's Composite Stock Price Index (IHSG)

77%

Monthly speed of adjustment (ECT) toward long-run equilibrium disequilibrium corrected in ~1.3 months

$R^2 = 0.60$

Share of monthly IHSG variation explained by the model

THE EMPIRICAL CONTEXT

IHSG moves at the intersection of global monetary policy and domestic investor behavior.

2020–2025



Study window: a period spanning pandemic recovery, an aggressive U.S. rate-hiking cycle, and renewed retail investor participation in Indonesia's capital market

68



Monthly observations used in estimation, after accounting for lag structure (effectively May 2020 – Dec 2025)

7 + 1



Independent variables capturing trading activity, an external shock (the Fed Funds Rate), market sentiment, and one exogenous control (MSCI dummy)

Why it matters: understanding whether IHSG movements are driven more by external shocks and foreign capital, or by domestic trading mechanics, has direct implications for how investors and policymakers read market signals during periods of global monetary tightening.

DATA AND RESEARCH DESIGN

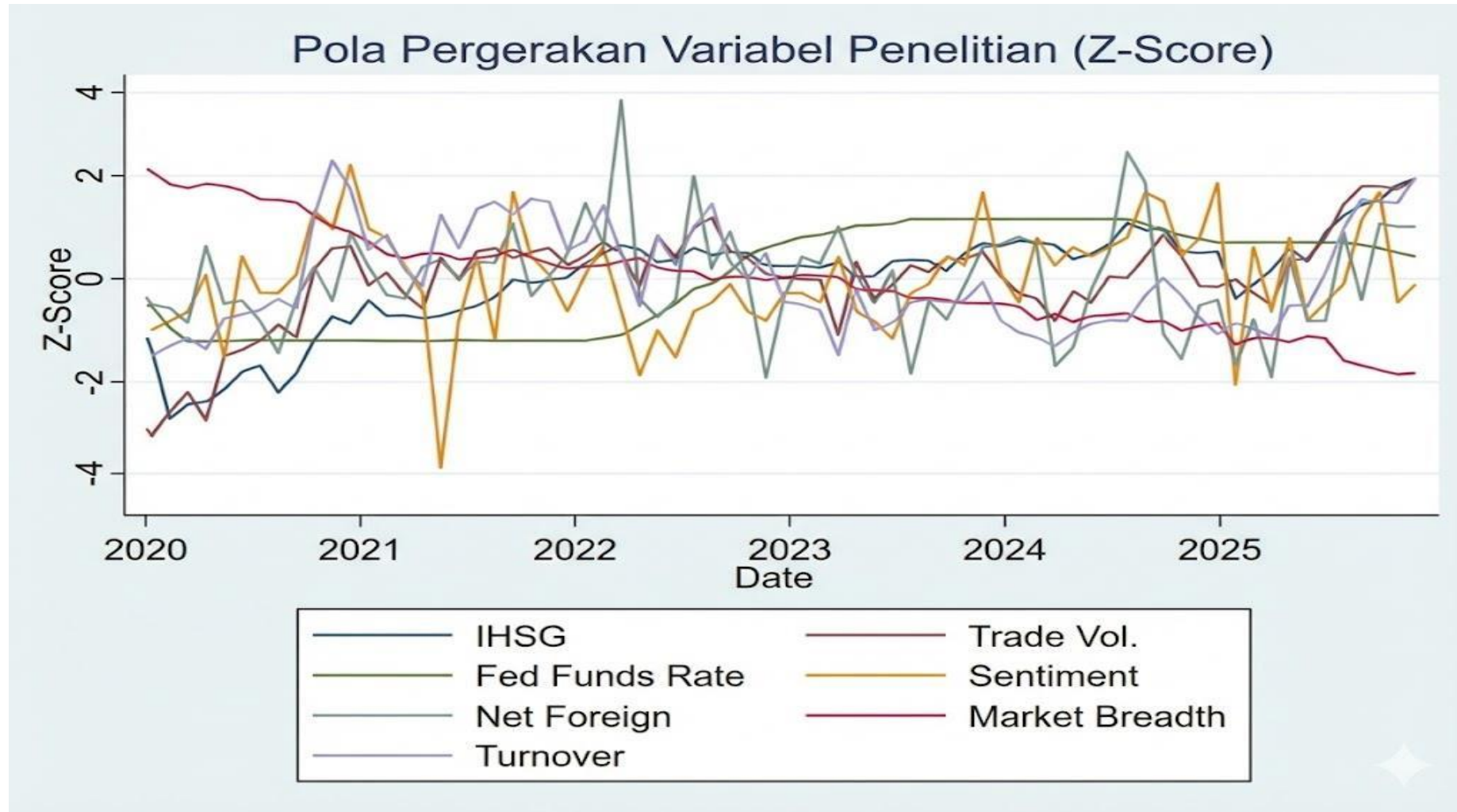
One outcome, six drivers, one exogenous control.

Role	Variable	Measure
Outcome (Y)	IHSG	Log of Indonesia's Composite Stock Price Index
Trading activity	TRADEV	Log of monthly stock trading volume
External shock	FEDFUNDS	U.S. Federal Funds Rate (%)
Market sentiment	SENTIMENT	Market sentiment index
Capital flow	NETFOREIGN	Net foreign capital flow (IDR trillion)
Trading activity	BREADTH	Market breadth (advancing vs. declining stocks)
Trading activity	TURNOVER	Share turnover ratio
Exogenous control	MSCI dummy	MSCI index rebalancing dummy

Monthly data, January 2020–December 2025 · Estimated with ARDL Bounds Testing (Pesaran, Shin & Smith, 2001) in Stata MP 17

DESCRIPTIVE ANALYSIS

Every series shows visible drift a first sign that formal stationarity testing is needed.



Standardized (Z-score) monthly movements of all study variables, 2020–2025. The visible trends and structural shifts particularly in IHSG, Fed Funds Rate, and Net Foreign flow motivate the unit root testing that follows.

A four stage ARDL - ECM Bounds Testing pipeline.

1 UNIT ROOT TESTING

Augmented Dickey-Fuller tests confirm variables are a mix of $I(0)$ and $I(1)$ the precondition for a valid ARDL bounds test.

2 LAG SELECTION & ESTIMATION

Optimal lag structure chosen via Akaike Information Criterion, selecting ARDL(1,1,1,0,4,3,0) as the most parsimonious specification.

3 COINTEGRATION (BOUNDS TEST)

F- and t-bounds tests (Pesaran, Shin & Smith, 2001) confirm a long-run equilibrium relationship among the variables.

4 DIAGNOSTICS & STABILITY

Normality, multicollinearity, heteroskedasticity, autocorrelation, and CUSUM/CUSUMQ stability tests validate the model.

STAGE 1 : UNIT ROOT TESTING

A mixed order of integration exactly what ARDL is built for.

Variable	Level (ADF)	1st Diff. (ADF)	Order
IHSG (ln)	−0.708 (n.s.)	−7.823 ***	I(1)
TRADEV (ln)	−2.532 (n.s.)	−11.660 ***	I(1)
FEDFUNDS	−0.486 (n.s.)	−3.440 ***	I(1)
SENTIMENT	−6.298 ***	−13.973 ***	I(0)
NETFOREIGN	−6.170 ***	−11.951 ***	I(0)
BREADTH	−1.455 (n.s.)	−8.738 ***	I(1)
TURNOVER	−2.410 (n.s.)	−9.799 ***	I(1)

NO I(2) VARIABLES

A mix of I(0) and I(1) variables, with none integrated at order two satisfying the precondition for a valid ARDL bounds test.

*** significant at 1% (MacKinnon, 1996 critical values). n.s. = not significant at level.

STAGE 2 : LAG SELECTION

ARDL(1, 1, 1, 0, 4, 3, 0) - chosen by AIC, not by assumption.

Variable	Role	Selected Lag
IHSG (ln)	Dependent	1
TRADEV (ln)	Independent	1
FEDFUNDS	Independent	1
SENTIMENT	Independent	0 (contemporaneous)
NETFOREIGN	Independent	4
BREADTH	Independent	3
TURNOVER	Independent	0 (contemporaneous)

AIC = -2.862223

N = 68 monthly observations (May 2020–Dec 2025)

A lag of 4 on Net Foreign flow means its effect on IHSG continues to unfold up to four months later while sentiment and turnover act contemporaneously, within the same month.

4 observations were dropped from the initial 72-month sample to accommodate the maximum lag of 4 on Net Foreign flow.

COINTEGRATION TEST

Both bounds tests confirm a long-run equilibrium relationship.

F-BOUNDS TEST

5.741

Exceeds the upper bound $I(1)$ of 5.200 even at the 1% significance level.

H_0 rejected cointegration exists

T-BOUNDS TEST

-5.733

More negative than the upper bound $I(1)$ critical value of -5.143 at the 1% level.

H_0 rejected cointegration exists

Both criteria are satisfied: IHSG and its determinants share a genuine long-run equilibrium relationship, validating the ARDL framework for further analysis.



LONG-RUN ESTIMATES

Net foreign flow is the dominant long-run driver of IHSG

Variable	Coefficient	t-stat	Sig.
TRADEV	0.0846	1.91	10%
FEDFUNDS	0.0136	2.72	1%
SENTIMENT	0.0022	2.20	5%
NETFOREIGN	0.0054	6.00	1%
BREADTH	-0.0001	-2.97	1%
TURNOVER	0.5248	0.55	n.s.

Joint significance: $F = 131.50 > F\text{-table } 2.29$ — all long-run variables are jointly significant ($p < 0.05$).

$$ECT = -0.7700$$

Negative and significant at 1% a valid error-correction mechanism.

77% of any deviation from long-run equilibrium is corrected within a single month equilibrium restored in ~1.3 months.



SHORT-RUN DYNAMICS

Net foreign flow acts fast and partially reverses within months.

Variable	Coeff.	t-stat	Sig.
D.TRADEV	0.0280	0.82	n.s.
D.FEDFUNDS	0.0603	2.37	5%
D.SENTIMENT	0.0017	2.43	5%
D.NETFOREIGN	0.0015	3.75	1%
D.NETFOREIGN (−1)	−0.0017	−2.43	5%
D.NETFOREIGN (−2)	−0.0013	−2.60	5%
D.BREADTH (−1)	0.0001	2.50	5%
D.TURNOVER	0.4041	0.56	n.s.

READING THE PATTERN

Net foreign flow has an immediate positive effect, but this partially reverses at 1- and 2-month lags consistent with short-term profit-taking after inflow surges.

Trading volume and turnover are not significant in the short run domestic trading mechanics matter more for long-run levels than month-to-month moves.



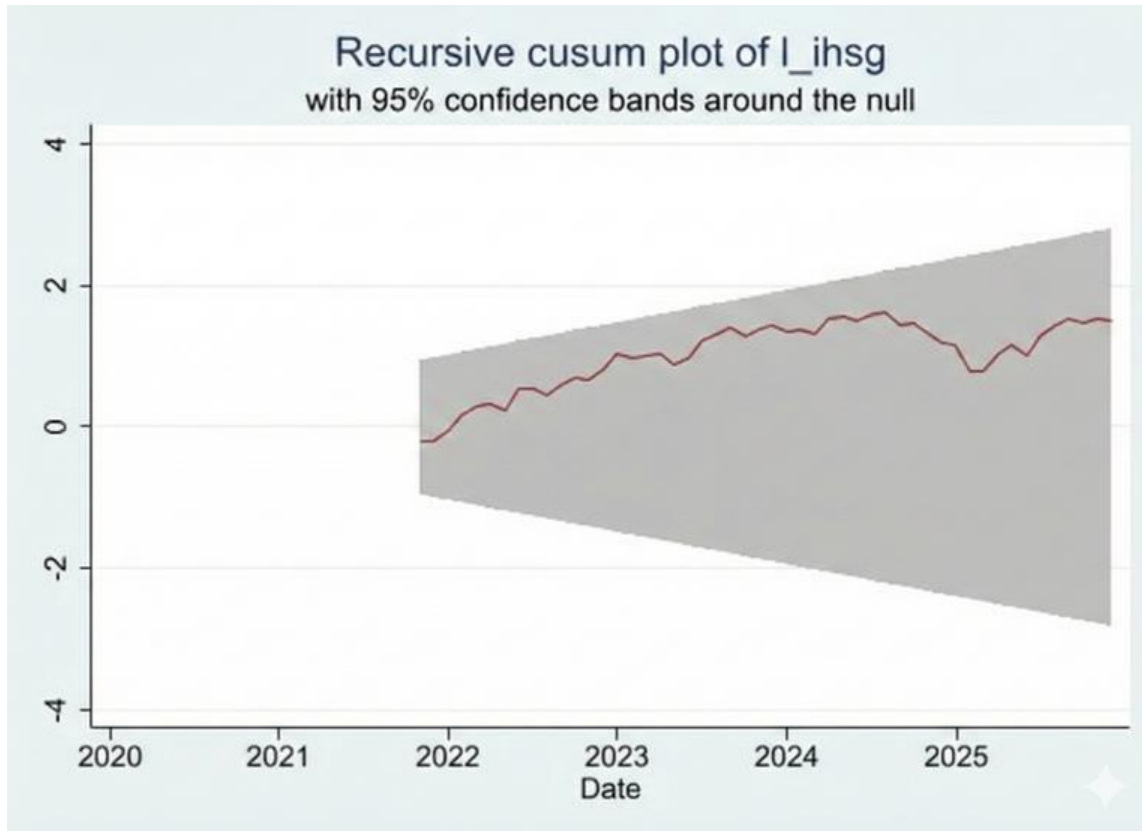
Every pairwise correlation stays below the 0.80 danger threshold.

	TRADEV	FEDFUNDS	SENTIMENT	NET-FOREIGN	BREADTH	TURNOVER	MSCI
TRADEV	1.000	0.239	0.168	0.265	−0.694	0.716	−0.009
FEDFUNDS	0.239	1.000	0.152	−0.104	−0.741	−0.419	0.011
SENTIMENT	0.168	0.152	1.000	0.187	−0.177	0.120	−0.079
NETFOREIGN	0.265	−0.104	0.187	1.000	−0.038	0.297	0.080
BREADTH	−0.694	−0.741	−0.177	−0.038	1.000	−0.090	−0.010
TURNOVER	0.716	−0.419	0.120	0.297	−0.090	1.000	−0.040
MSCI	−0.009	0.011	−0.079	0.080	−0.010	−0.040	1.000

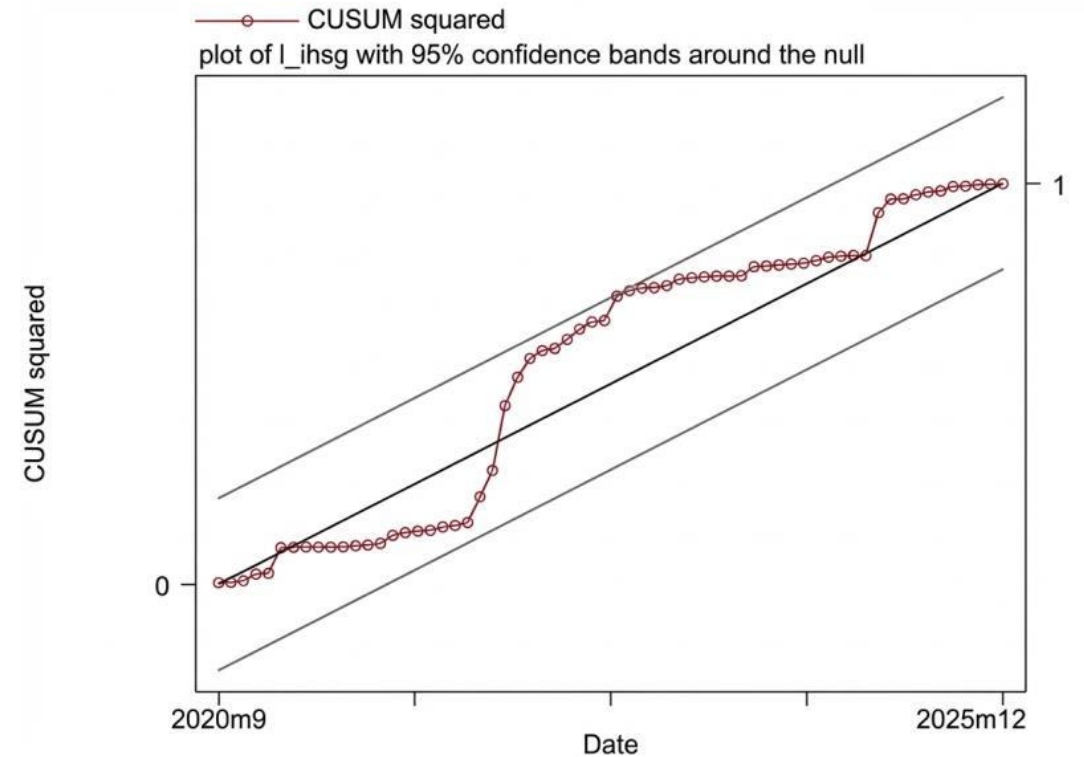
Highest correlations: $BREADTH \times FEDFUNDS = -0.741$, and $TURNOVER \times TRADEV = 0.716$ both still within the tolerable range (Gujarati & Porter, 2009), so all seven variables are retained in the model.

DIAGNOSTICS DETAIL : STRUCTURAL STABILITY

CUSUM and CUSUM-of-Squares stay within the 5% critical bounds.



Recursive CUSUM plot stays inside the confidence bands throughout



CUSUM-of-Squares plot no breach of the critical bands

MODEL DIAGNOSTICS

The model passes every classical assumption test.

NORMALITY

$\chi^2 = 1.71, p = 0.426$

✓ **Residuals normal**

MULTICOLLINEARITY

$\max |r| = 0.74$

✓ **Below 0.80 threshold**

HETEROSKEDASTICITY

White test, $p = 0.443$

✓ **Homoscedastic**

AUTOCORRELATION

Breusch–Godfrey, $p = 0.076$

✓ **No autocorrelation**

STRUCTURAL STABILITY

CUSUM / CUSUMQ

✓ **Within 5% bounds**

GOODNESS OF FIT

$R^2 = 0.5957$

✓ **59.57% variance explained**

KEY FINDINGS

Foreign capital flows and market sentiment dominate IHSG's dynamics and the market corrects itself fast.

77%/mo

Speed of adjustment toward long-run equilibrium disequilibrium is corrected within roughly 1.3 months

Net Foreign Flow

The single most dominant long-run determinant (highest t-statistic among all covariates)

Breadth: negative

A rare finding narrow, concentrated rallies are associated with weaker long-run index performance

Rifqi Firmawan

Economic Graduate